

Fixed-Polylogarithmic Natural-Density Descent for the Collatz Map

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Abstract

Let

$$T(n) = \begin{cases} n/2, & n \equiv 0 \pmod{2}, \\ (3n+1)/2, & n \equiv 1 \pmod{2}, \end{cases} \quad T_{\min}(n) = \min_{k \geq 0} T^k(n).$$

Put

$$A_{\text{FP}} = \frac{1}{2(1 - H_2(\log_3 2))} = 9.9911133419 \dots, \quad c_* = \frac{2}{\log(4/3)} = 6.9521189935 \dots,$$

where H_2 is binary entropy. For every fixed $A > A_{\text{FP}}$, $c > c_*$, and $\beta > 0$, we prove that all but $O(X / (\log X)^\kappa)$ integers $n \leq X$ possess a shortcut-Collatz iterate, before $c \log n$ steps, satisfying

$$T^k(n) \leq C(\log n)^A, \quad \max_{0 \leq j \leq k} T^j(n) \leq n^{1+\beta}.$$

The constants $C, \kappa > 0$ may depend on the displayed fixed parameters. The endpoint $A = A_{\text{FP}}$ is not asserted. At the weaker target $\exp((\log n)^{1-\delta})$, the same method gives the sharper exceptional count

$$O_{\delta,c,\beta}(X \exp(-c_{\delta,c,\beta}(\log X)^{1-\delta}))$$

for every fixed $0 < \delta < 1$, $c > c_*$, and $\beta > 0$; the same witnesses give every raw Collatz clock constant greater than $3 / \log(4/3)$.

The proof uses exact parity-cube counting only to certify an all-prefix orbit barrier. It then proceeds deterministically. Strictly nested thresholds turn every later certification failure into a direct first passage from the original shell, while shrinking high-rank barriers confine its feasible cumulative times to $O(\sqrt{M \log M})$ values. Transport over this smaller time support replaces a linear horizon loss by a square-root loss and yields the displayed exponent. The result is an almost-all statement and makes no claim for every starting value.

1. Introduction and main results

We ask how small a value can be forced along the Collatz orbit of almost every initial integer in ordinary natural density, while retaining a logarithmic witnessing time and a bound on the orbit before that witness. This separates typical descent from universal termination; no result below applies to every starting value.

We work throughout with the shortcut Collatz map T displayed in the abstract. All unqualified logarithms are natural, $\log_2$ denotes the base-two logarithm, and $\mathbb{N} = \{1, 2, 3, \dots\}$. Put

$$a_0 = \frac{\log_2 3}{2}. \quad (1.1)$$

Thus $a_0 - 1$ is the mean binary logarithmic multiplier per shortcut step under the exact uniform parity coding used below.

Define binary entropy by

$$H_2(p) = -p \log_2 p - (1 - p) \log_2(1 - p),$$

and put

$$p_* = \log_3 2, \quad A_{FP} = \frac{1}{2(1 - H_2(p_*))}, \quad c_* = \frac{2}{\log(4/3)}. \quad (1.2)$$

Theorem 1.1 (optimized fixed-polylogarithmic descent)

For every fixed

$$A > A_{FP}, \quad c > c_*, \quad \beta > 0, \quad (1.3)$$

there are constants $C, \kappa, X_0 > 0$ with the following property. For every $X \geq X_0$, let $E_{A,c,\beta,C}(X)$ be the set of integers $1 \leq n \leq X$ for which no integer $0 \leq k < c \log n$ simultaneously satisfies

$$T^k(n) \leq C(\log n)^A, \quad \max_{0 \leq j \leq k} T^j(n) \leq n^{1+\beta}.$$

Then

$$\# E_{A,c,\beta,C}(X) \leq C \frac{X}{(\log X)^\kappa}. \quad (1.4)$$

In particular, $T_{\min}(n) \leq C(\log n)^A$ on a set of natural density one. The endpoint $A = A_{FP}$ is not asserted.

Since $(\log n)^A = o(n)$, the displayed landing is a genuine descent for all sufficiently large n . Neither the constant C nor the onset of this crossover is made effective.

Theorem 1.2 (endpoint-rate stretched-logarithmic descent)

For every fixed $0 < \delta < 1$, $c > c_*$, and $\beta > 0$, there are constants $C_{\delta,c,\beta}, c_{\delta,c,\beta}, X_{\delta,c,\beta} > 0$ with the following property. For every $X \geq X_{\delta,c,\beta}$, let $E_{\delta,c,\beta}(X)$ be the set of integers $1 \leq n \leq X$ for which no integer $0 \leq k < c \log n$ simultaneously satisfies

$$T^k(n) \leq \exp((\log n)^{1-\delta}), \quad \max_{0 \leq j \leq k} T^j(n) \leq n^{1+\beta}.$$

Then

$$\# E_{\delta,c,\beta}(X) \leq C_{\delta,c,\beta} X \exp(-c_{\delta,c,\beta}(\log X)^{1-\delta}). \quad (1.5)$$

The endpoint $\delta = 1$ is not asserted.

Corollary 1.3 (raw clock)

Put

$$\text{Col}(n) = \begin{cases} n/2, & n \equiv 0 \pmod{2}, \\ 3n+1, & n \equiv 1 \pmod{2}. \end{cases}$$

For every fixed $A > A_{\text{FP}}$, $\beta > 0$, and

$$c_{\text{raw}} > \frac{3}{\log(4/3)}, \quad (1.6)$$

there are constants $C, \kappa, X_0 > 0$ such that all but $CX/(\log X)^\kappa$ integers $n \leq X$, for $X \geq X_0$, possess an integer $\ell < c_{\text{raw}} \log n$ for which

$$\text{Col}^\ell(n) \leq C(\log n)^A, \quad \max_{0 \leq j \leq \ell} \text{Col}^j(n) \leq n^{1+\beta}.$$

The analogous statement with the target in Theorem 1.2 and its exceptional rate also holds. Since $3/\log(4/3) = 10.42817849 \dots$, the explicit constant 10.44 remains admissible.

Corollary 1.4 (fixed powers and the graded clock)

For every fixed $\alpha > 0$, the exceptional set for $T_{\min}(n) > n^\alpha$ is

$$O_{\alpha,\sigma}(X \exp(-c_{\alpha,\sigma}(\log X)^\sigma)) \quad (1.7)$$

for every fixed $0 < \sigma < 1$.

If $0 < \alpha < 1$, then for every fixed $\varepsilon > 0$ the same fixed-power target is also reached on a natural-density-one set before

$$\left(\frac{2(1-\alpha)}{\log(4/3)} + \varepsilon \right) \log n \quad (1.8)$$

shortcut steps. Thus the clock decreases continuously with the amount of logarithmic height that remains: for example, the limiting coefficient at $\alpha = 1/2$ is $1/\log(4/3) = 3.47605949 \dots$, rather than the full descent coefficient c_* .

Relation to previous almost-all results

The preceding endpoint-transport preprint [7] proves the range $0 < \delta < 0.251245530155874 \dots$ by a fixed-time endpoint-fiber/Rényi estimate followed by endpoint iteration. The present proof is independent of that theorem. It replaces fixed-time endpoint transport by all-prefix certification, first-passage reversal, nested direct re-certification, and support-sensitive time aggregation.

The main obstruction is not the contraction of one typical block, but the transport of sparse certification failures through a generated sequence of landings. A direct aggregation over the available time tags incurs a linear horizon loss. The support theorem below compresses those tags to $O(\sqrt{M \log M})$ possibilities; this linear-to-square-root reduction is the quantitative pivot of the paper.

The theorem should be read along several independent axes: density notion, target scale, clock, and exceptional rate. The closest comparisons are:

Result	Status and proof relationship	Density	Target reached for almost all starts	Clock retained in the stated result	Quantitative exception
Korec [3]	published	natural	n^θ , every fixed $\theta > a_0$	not part of the cited theorem	not used here
Inselmann [2]	preprint	natural	n^ε , every fixed $\varepsilon > 0$	$2 \log n / \log(4/3)$ shortcut steps	density convergence
Tao [5]	published; original Syracuse/3-adic first-passage framework	logarithmic	every $f(n) \rightarrow \infty$	no single global clock in the headline theorem	logarithmic-density estimate
Mazur [4]	manuscript with a pinned formal artifact; Tao-to-natural-density bridge	natural	every $f(n) \rightarrow \infty$	$< 436 \log n$ raw steps	fixed-target logarithmic rate
Allikvere [1]	unrefereed preprint; Tao-to-natural-density bridge	natural	every $f(n) \rightarrow \infty$	$< 12 \log n$ raw steps	$O((\log N_0)^{-1/29} + X^{-1/2000})$ for a fixed target
This paper	research draft; independent first-passage transport chain	natural	$C(\log n)^A$, every fixed $A > A_{FP}$; also $\exp((\log n)^{1-\delta})$ with a stronger rate	every shortcut constant $> c_*$; every raw constant $> 3 / \log(4/3)$; graded clock (1.8) for fixed powers	$O(X / (\log X)^\alpha)$ at the poly-log target; $O(X e^{-c(\log X)^{1-\delta}})$ at the stretched-log target

The fixed-polylogarithmic target is smaller than every fixed power and every fixed stretched-logarithmic target of the form in Theorem 1.2. It remains weaker than an arbitrary diverging function. Thus the present theorem does not supersede the arbitrary-threshold conclusions of [1] or [4], and the clock scale already appears in [2]. The comparison is deliberately coordinatewise: target scale, density notion, exceptional rate, and clock are not collapsed into one ordering.

The two recent arbitrary-diverging natural-density manuscripts are bridge results from Tao's first-passage program. Allikvere conditions Tao's Syracuse/Fourier inputs on the total valuation and supplies a uniform-measure transfer. Mazur starts from the same logarithmic-density frontier and adds a quantitative phase-gap and two-adic lift. The proof below does not import Tao's renewal or Fourier-mixing estimates: its probabilistic input is exact Boolean-cube counting, followed by deterministic first-passage transport.

The probabilistic input below is confined to exact parity-cube counting used to certify an all-prefix barrier. From first-passage reversal onward, the transport and re-certification argument is deterministic.

The proof has two sharply separated parts and six load-bearing steps.

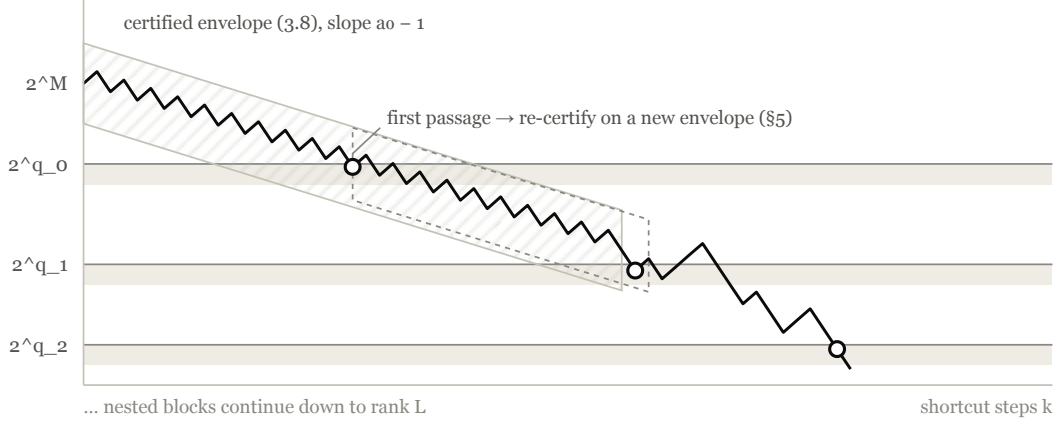
Exact probabilistic certification.

1. The parity-vector bijection turns a complete dyadic shell into the uniform Boolean cube.
2. An adjustable maximal Boolean barrier gives an entropy-sharp shell exceptional rate and a certified all-prefix orbit envelope.

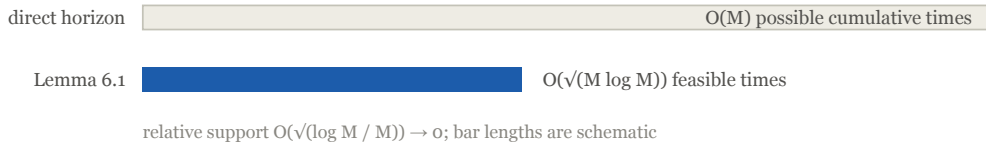
Deterministic Collatz transport and assembly.

3. First-passage reversal gives a loss-filtered tagged-fiber bound for every target subset of a landing shell.
4. Strictly decreasing threshold ranks identify each recursively generated landing with a direct first passage from the original source shell.
5. Rank-dependent high barriers confine every cumulative first-bad time to a common support of size $O(\sqrt{M \log M})$.
6. Support-sensitive transport and a high/low parameter choice separate the leading clock from the low-rank entropy rate and yield A_{FP} .

(a) certified descent and nested first passages — a real orbit in I_M , $M = 40$
orbit value



(b) the time-tag compression used by the transport argument (*orders only*)



The proof architecture. Panel (a) follows an actual shortcut-Collatz orbit inside one certified all-prefix envelope and marks the nested first-passage landings at which the deterministic argument re-certifies. Panel (b) records the proved order reduction in the number of feasible cumulative time tags; bar lengths are schematic and do not encode the unspecified constant in Lemma 6.1.

2. Density, parity, and affine iterates

The argument uses three common coordinates: a shell-to-prefix summation rule, exact parity coding on each dyadic shell, and an affine formula for the iterate. This section establishes them for the certification and transport arguments in Section 3 and Section 4.

For $S \subseteq \mathbb{N}$, put

$$B_S(X) = \# \{1 \leq n \leq X : n \notin S\}.$$

We call S (C, D) -**dense** if

$$B_S(X) \leq CX^{1-D} \quad (X \geq 1), \quad (2.1)$$

where $C > 0$ and $0 < D \leq 1$. Every such set has natural density one.

We use the following varying-rate shell summation.

Lemma 2.1 (varying-rate dyadic summation)

Let $0 < \sigma \leq 1$, $A, \gamma > 0$, and suppose that, for all sufficiently large M ,

$$\#(E \cap [2^M, 2^{M+1})) \leq Ae^{-\gamma(M+4)^\sigma} 2^M. \quad (2.2)$$

Then there are $\gamma' > 0$ and X_0 such that

$$\#(E \cap [1, X]) \leq (2A + 1)Xe^{-\gamma'(\log X)^\sigma} \quad (X \geq X_0). \quad (2.3)$$

Proof

Let $2^J \leq X < 2^{J+1}$. The shells below $J/2$ contain fewer than $X^{1/2}$ integers. On the remaining shells,

$$M + 4 \geq \frac{\log X}{4 \log 2}.$$

Thus their total contribution is at most

$$2AX \exp\left(-\frac{\gamma}{(4 \log 2)^\sigma} (\log X)^\sigma\right).$$

Choose

$$\gamma' = \min\left\{\frac{\gamma}{(4 \log 2)^\sigma}, \frac{1}{2}\right\}.$$

For all sufficiently large X , one has $X^{1/2} \leq X e^{-\gamma' (\log X)^\sigma}$. Adding the early and late contributions proves (2.3).

□

For $n \geq 1$, define

$$p_i(n) = 1_{\{T^i(n) \text{ odd}\}}, \quad s_k(n) = \sum_{i=0}^{k-1} p_i(n). \quad (2.4)$$

The exact parity coding below is classical in the stopping-time literature; see, for example, Terras [6]. Its short proof is included because no external result is needed by the argument.

Proposition 2.2 (parity-vector bijection)

For every $M \geq 0$,

$$n \bmod 2^M \mapsto (p_0(n), \dots, p_{M-1}(n)) \quad (2.5)$$

is a bijection from $\mathbb{Z}/2^M\mathbb{Z}$ to $\{0, 1\}^M$.

Proof

The relation

$$n \equiv n' \pmod{2^{k+1}} \implies T(n) \equiv T(n') \pmod{2^k}$$

shows inductively that the first M parity bits depend only on $n \bmod 2^M$. For the converse, induct on the word length. Suppose the first bit is e , and use the induction hypothesis to recover the residue $m = T(n) \bmod 2^k$ from the remaining k bits. If $e = 0$, the unique compatible residue is

$$n \equiv 2m \pmod{2^{k+1}}.$$

If $e = 1$, it is

$$n \equiv (2m - 1)3^{-1} \pmod{2^{k+1}},$$

where 3 is a unit modulo 2^{k+1} . The first residue is even, the second is odd, and in either case applying T recovers $m \bmod 2^k$. Thus every word has exactly one preimage residue. □

Proposition 2.3 (exact affine iterate)

For every $n \geq 1$ and $k \geq 0$,

$$T^k(n) = \frac{3^{s_k(n)}}{2^k} n + \sum_{i=0}^{k-1} \frac{p_i(n) 3^{s_k(n) - s_{i+1}(n)}}{2^{k-i}}. \quad (2.6)$$

Proof

The assertion is trivial at $k = 0$. Passing from k to $k + 1$, an even letter divides every existing term by two. An odd letter multiplies every existing term by $3/2$ and adds $1/2$. This is exactly the displayed update. $\square$

3. A dense maximal-barrier set

The parity bijection identifies a complete dyadic shell with the Boolean cube. We now discard the sparse words whose prefixes deviate too far from their mean and obtain the pathwise orbit envelope needed to guarantee the first passages used later.

For a parity word of length M , put

$$Y_k = 2s_k - k, \quad H_M = \max_{0 \leq k \leq M} s_k - \frac{k}{2} = \frac{1}{2} \max_{0 \leq k \leq M} |Y_k|. \quad (3.1)$$

For $0 \leq t < 1/2$, put

$$I(t) = D\left(\frac{1}{2} + t, \frac{1}{2}\right) = \left(\frac{1}{2} + t\right) \log(1 + 2t) + \left(\frac{1}{2} - t\right) \log(1 - 2t). \quad (3.2)$$

Lemma 3.1 (entropy-sharp maximal Boolean-walk bound)

If $M \geq 1$ and $0 \leq h < M/2$, then

$$2^{-M} \# \{w \in \{0, 1\}^M : H_M(w) > h\} \leq 2 \exp(-M I(h/M)). \quad (3.3)$$

In particular, the right side is at most $2 \exp(-2h^2/M)$.

Proof

Fix $\theta > 0$ and stop the Boolean tree at the first prefix u with $|Y(u)| > 2h$. For a frontier node of depth k , set

$$\Phi_\theta(u) = 2^{-k} \cosh(\theta Y(u)) (\cosh \theta)^{M-k}.$$

The identity

$$\cosh(\theta(y-1)) + \cosh(\theta(y+1)) = 2 \cosh(\theta y) \cosh \theta$$

shows that replacing an unexpanded node by its two children preserves the total potential. The frontier formed by first-crossing prefixes and noncrossing depth- M words therefore has total $(\cosh \theta)^M$. Each crossing prefix contributes at least $2^{-|u|} \cosh(2\theta h)$, whence

$$\Pr(H_M > h) \leq 2 \exp(M \log \cosh \theta - 2\theta h). \quad (3.4)$$

Take $u = 2h/M$ and $\theta = \operatorname{arctanh} u$. The elementary Legendre identity

$$u\theta - \log \cosh \theta = D\left(\frac{1+u}{2}, \frac{1}{2}\right) = I(h/M)$$

proves (3.3). Pinsker's inequality in this binary case, or the direct convexity estimate $I(t) \geq 2t^2$, gives the final assertion. $\square$

Set

$$\varrho := \frac{\sqrt{3}}{2} = 2^{a_0-1}.$$

Define

$$r_k(n) = \sum_{i=0}^{k-1} \frac{p_i(n)}{3^{s_{i+1}(n)} 2^{k-i}}, \quad d_k = s_k - \frac{k}{2}. \quad (3.5)$$

By Proposition 2.3,

$$T^k(n) = \varrho^k n 3^{d_k} + (r_k(n) 3^{k/2}) 3^{d_k}. \quad (3.6)$$

Lemma 3.2 (uniform affine correction)

If $H_M \leq h$, then for every $1 \leq k \leq M$,

$$r_k(n) 3^{k/2} \leq (2 + \sqrt{3})(1 - \varrho^k) 3^h < (2 + \sqrt{3}) 3^h. \quad (3.7)$$

Proof

For $0 \leq i < k$, the barrier gives $s_{i+1} \geq (i+1)/2 - h$. Hence

$$r_k 3^{k/2} \leq 3^h \sum_{i=0}^{k-1} \frac{3^{(k-i-1)/2}}{2^{k-i}} = \frac{3^h}{2} \sum_{j=0}^{k-1} \varrho^j,$$

which is (3.7). $\square$

For $0 < \eta \leq 1$, let W_η be the set of integers n such that

$$\varrho^k n^{1-\eta} \leq T^k(n) \leq \varrho^k n^{1+\eta} \quad (0 \leq k \leq \lfloor \log_2 n \rfloor). \quad (3.8)$$

Define

$$b_{\text{ent}}(\eta) = I\left(\frac{\eta}{\log_2 3}\right). \quad (3.9)$$

Proposition 3.3 (entropy-sharp density of the maximal-barrier set)

Let $0 < \eta < 1 - a_0$. For every fixed $0 < c < b_{\text{ent}}(\eta)$, there is $K_{\eta,c} > 0$ such that

$$\frac{\#(W_\eta^c \cap I_M)}{2^M} \leq K_{\eta,c} e^{-cM} \quad (M \geq 0). \quad (3.10)$$

Consequently W_η is $(K_{\eta,c}', c/\log 2)$ -dense after increasing the fixed constant.

Proof

Put $L_3 = \log_2 3$. Since $c < b_{\text{ent}}(\eta)$, choose a fixed $0 < \lambda < 1$ with

$$c < I\left(\frac{\lambda\eta}{L_3}\right). \quad (3.11)$$

On I_M , take

$$h = \frac{\lambda \eta M}{L_3}. \quad (3.12)$$

Assume $H_M \leq h$. Since $3^h = 2^{\lambda \eta M} \leq n^\eta$, the multiplicative term in (3.6) is at least $Q^k n^{1-\eta}$ and at most

$$2^{-(1-\lambda)\eta M} Q^k n^{1+\eta}. \quad (3.13)$$

Lemma 3.2 bounds the additive term by

$$(2 + \sqrt{3})3^{2h} = (2 + \sqrt{3})2^{2\lambda \eta M}. \quad (3.14)$$

The unused part of the upper envelope is at least

$$(1 - 2^{-(1-\lambda)\eta M}) Q^M 2^{(1+\eta)M}. \quad (3.15)$$

Its binary exponent exceeds that of (3.14), because

$$a_0 + \eta - 2\lambda \eta \geq a_0 - \eta > 0. \quad (3.16)$$

Thus (3.8) holds for every $k \leq M$ once M is sufficiently large.

Proposition 2.2 and Lemma 3.1 now give

$$\frac{\#(W_\eta^c \cap I_M)}{2^M} \leq 2 \exp[-M I(\frac{\lambda \eta}{L_3})] \leq 2e^{-cM}$$

on all sufficiently large shells. Enlarge $K_{\eta,c}$ to absorb the finite startup. Summing the resulting geometric shell series proves the global density assertion. $\square$

4. First-passage linear transport

The barrier controls the retained sources, but re-certification also requires transporting an arbitrary bad landing set back to the original shell. First-passage reversal and loss-filtered fibers provide that bridge.

For real $Y > 1$, define

$$\tau_Y(n) = \min\{h \geq 0 : T^h(n) \leq Y\} \quad (4.1)$$

when the set is nonempty.

Lemma 4.1 (first-passage band and reverse product)

Let $1 < Y < 2^M$, let $n \in I_M$, and suppose $\tau_Y(n) = h \geq 1$. Write $x_j = T^j(n)$, $y = x_h$, and let s be the number of odd values among $x_0, \dots, x_{h-1}$. Then

$$\frac{Y}{2} < y \leq Y \quad (4.2)$$

and

$$n = \frac{2^h y}{3^s} \prod_{\substack{0 \leq j < h \\ x_j \text{ odd}}} \left(1 - \frac{1}{2x_{j+1}}\right). \quad (4.3)$$

Consequently, whenever $h < 2Y$,

$$\left(1 - \frac{h}{2Y}\right) \frac{2^h y}{3^s} \leq n \leq \frac{2^h y}{3^s}. \quad (4.4)$$

Proof

The final crossing cannot be odd: otherwise $x_h = (3x_{h-1} + 1)/2 > x_{h-1} > Y$. Hence $x_{h-1} = 2y$, proving (4.2). Reversing an even step gives $x_j = 2x_{j+1}$, while reversing an odd step gives

$$x_j = \frac{2x_{j+1} - 1}{3} = \frac{2}{3}x_{j+1} \left(1 - \frac{1}{2x_{j+1}}\right).$$

Multiplication proves (4.3). Every odd factor occurs before the final crossing, so its following state is greater than Y . Therefore $\prod_i (1 - u_i) \geq 1 - \sum_i u_i$ gives a lower product bound $1 - s/(2Y) \geq 1 - h/(2Y)$; the upper bound is one. $\square$

Write the reverse product in (4.3) as $\prod_{j < h} (1 - u_j)$, where

$$u_j(n) = \begin{cases} \frac{1}{2T^{j+1}(n)}, & T^j(n) \text{ odd}, \\ 0, & T^j(n) \text{ even}, \end{cases} \quad E_Y(n) = Y \sum_{j=0}^{h-1} u_j(n). \quad (4.5)$$

The elementary product inequality gives

$$1 - \frac{E_Y(n)}{Y} \leq \prod_{j=0}^{h-1} (1 - u_j(n)) \leq 1. \quad (4.6)$$

Lemma 4.2 (loss-filtered tagged fibers)

Let $D \geq 0$, and suppose

$$\frac{D}{Y} \leq \frac{1}{3}. \quad (4.7)$$

For a fixed tag (h, y) , the number of sources $n \in I_M$ satisfying

$$\tau_Y(n) = h, \quad T^h(n) = y, \quad E_Y(n) \leq D$$

is at most

$$1 + 3D \frac{2^M}{Y}. \quad (4.8)$$

Proof

Put $P(n) = \prod_{j < h} (1 - u_j(n))$. By Lemma 4.1, if two such sources had odd counts $s_1 < s_2$, put $A_i = 2^h y / 3^{s_i}$. Then $A_1 \geq 3A_2$, while (4.3) and (4.6) give

$$n_1 = A_1 P(n_1) \geq 3A_2 \left(1 - \frac{D}{Y}\right) \geq 2A_2 \geq 2n_2.$$

This is impossible in the half-open shell I_M . The argument uses only positive products and introduces no division by a reverse product. Fix the common odd count and put $A = 2^h y / 3^s$. All sources lie in $[(1 - D/Y)A, A]$. Nonemptiness, (4.7), and $n < 2^{M+1}$ imply $A < (3/2)2^{M+1}$, so this interval has length less than

$$3D \frac{2^M}{Y}.$$

This proves (4.8). $\square$

Proposition 4.3 (loss-filtered target transport)

Put $J_Y = (Y/2, Y] \cap \mathbb{N}$. Let $1 < Y < 2^M$. Under (4.7), every $B \subseteq J_Y$ satisfies the exact bound

$$\# \{ n \in I_M : \begin{array}{l} \tau_Y(n) \leq H, \quad T^{\tau_Y(n)}(n) \in B, \\ E_Y(n) \leq D \end{array} \} \leq H \left(1 + 3D \frac{2^M}{Y} \right) |B|. \quad (4.9)$$

Consequently,

$$\# \{ n \in I_M : \begin{array}{l} \tau_Y(n) \leq H, \quad T^{\tau_Y(n)}(n) \in B, \\ E_Y(n) \leq D \end{array} \} \leq H(1 + 3D) \frac{2^M}{Y} |B|. \quad (4.10)$$

The estimate remains valid after arbitrary source restriction.

Proof

Since $Y < 2^M$, every passage time is positive. Sum Lemma 4.2 over the at most $H|B|$ tags to obtain (4.9). The contribution $H|B|$ is at most $H(2^M/Y)|B|$, which gives (4.10). $\square$

5. Nested first-passage re-certification

The transport theorem applies directly to a first passage from the original shell. The nested construction below shows that every first certification failure has exactly this form and controls the reverse loss accumulated before it.

Fix

$$a_0 < r < 1, \quad 0 < \eta < r - a_0, \quad 0 < c_\eta < b_{\text{ent}}(\eta). \quad (5.1)$$

Increase a fixed startup rank $M_0 = M_0(r, \eta, c_\eta)$ whenever needed below. If $x \in W_\eta \cap I_m$, put $q = \lfloor r m \rfloor$. For all $m \geq M_0$,

$$T^m(x) \leq Q^m x^{1+\eta} < 2^{(a_0+\eta)m+1+\eta} \leq 2^q. \quad (5.2)$$

Thus $\tau_{2^q}(x) \leq m$, and (3.8) bounds the whole block by $x^{1+\eta}$.

Fix integers $M \geq L \geq M_0$, and start with $n_0 = n \in I_M$. Put

$$m_i = \lfloor \log_2 n_i \rfloor, \quad t_0 = 0.$$

If $m_i < L$, stop successfully. If $m_i \geq L$ but $n_i \notin W_\eta$, stop with a certification failure. Otherwise define

$$q_i = \lfloor r m_i \rfloor, \quad h_i = \tau_{2^{q_i}}(n_i), \quad n_{i+1} = T^{h_i}(n_i), \quad t_{i+1} = t_i + h_i. \quad (5.3)$$

The first-passage band gives

$$2^{q_i-1} < n_{i+1} \leq 2^{q_i}, \quad m_{i+1} \in \{q_i - 1, q_i\}. \quad (5.4)$$

Since $q_i \leq m_i - 1$,

$$m_{i+1} \leq q_i \leq r m_i, \quad q_{i+1} \leq q_i - 1 \quad (5.5)$$

whenever the next block is executed. Consequently

$$t_i \leq \sum_{j < i} m_j \leq M \sum_{j < i} r^j < \frac{M}{1-r}. \quad (5.6)$$

Lemma 5.1 (nested direct first passage)

For every executed block i ,

$$t_{i+1} = \tau_{2^{q_i}}(n_0), \quad n_{i+1} = T^{\tau_{2^{q_i}}(n_0)}(n_0). \quad (5.7)$$

Proof

The assertion is the definition when $i = 0$. Inductively, every state before t_i exceeds the preceding threshold and hence the strictly smaller 2^{q_i} . At time t_i , one has $n_i \geq 2^{m_i} > 2^{q_i}$. The local block stays above 2^{q_i} until its end-point. Thus $t_i + h_i$ is exactly the first crossing of the new threshold by the original orbit. $\square$

For $q \geq 1$, put

$$J_q = (2^{q-1}, 2^q] \cap N, \quad B_q = W_\eta^c \cap J_q. \quad (5.8)$$

The interval J_q lies in I_{q-1} apart from its upper endpoint. Proposition 3.3 therefore gives

$$\frac{|B_q|}{2^q} \leq C_{\eta, c_\eta} (e^{-c_\eta q} + 2^{-q}). \quad (5.9)$$

Lemma 5.2 (rank-scaled reverse loss)

If blocks $0, \dots, i$ are executed before the first certification failure, then the direct first passage in (5.7) satisfies

$$E_{2^{q_i}}(n) < \frac{q_i + 2}{r}. \quad (5.10)$$

Proof

In block j , every state following an odd step is strictly larger than 2^{q_j} ; the final crossing is even. Therefore each reverse-loss increment is less than 2^{-q_j-1} . Since the block has at most m_j steps, rescaling all blocks to the final threshold gives

$$E_{2^{q_i}}(n) < \frac{1}{2} \sum_{j \leq i} m_j 2^{q_i - q_j}.$$

Now $m_j < (q_j + 1)/r$, and the distinct integer thresholds q_j are all at least q_i . Hence

$$E_{2^{q_i}}(n) < \frac{1}{2r} \sum_{d \geq 0} (q_i + d + 1) 2^{-d} = \frac{q_i + 2}{r}.$$

$\square$

Let $\text{Fail}_{M,L}$ denote the sources whose chain reaches a certification failure before it reaches rank below L .

Theorem 5.3 (optimized terminal profile)

Under (5.1), there is a fixed startup rank and a constant $C > 0$ such that, for all $M \geq L$ above that startup,

$$\frac{\# \text{ Fail}_{M,L}}{2^M} \leq C [e^{-c_\eta M} + M(L+1)(e^{-c_\eta L} + 2^{-L})]. \quad (5.11)$$

Every source outside this set has an integer $k < M/(1-r)$ such that

$$T^k(n) < 2^L, \quad \max_{0 \leq j \leq k} T^j(n) \leq n^{1+\eta}. \quad (5.12)$$

Proof

An initial failure has proportion $O(e^{-c_\eta M})$. Otherwise let the first failed certification be a generated landing in B_q . Every earlier block and the block producing that landing are certified, so Lemma 5.2 applies; the bad landing itself is not assumed certified. Lemma 5.1 makes it a direct first passage from I_M , at time at most $M/(1-r) + 1$.

Increase the startup rank until

$$\frac{q+2}{r2^q} \leq \frac{1}{3} \quad (5.13)$$

for every $q \geq L$. Proposition 4.3, with $D = (q+2)/r$, bounds the proportion of sources whose first failed landing has rank q by

$$CM(q+1)(e^{-c_\eta q} + 2^{-q}). \quad (5.14)$$

The two elementary tail estimates

$$\sum_{q \geq L} (q+1)e^{-c_\eta q} \ll (L+1)e^{-c_\eta L}, \quad \sum_{q \geq L} (q+1)2^{-q} \ll (L+1)2^{-L}$$

prove (5.11).

Outside the failure set, integer ranks strictly decrease until a state of rank below L is reached. Equations (5.2), (5.6), and the monotonic decrease of block endpoints prove (5.12). $\square$

6. Shrinking barriers and compressed time support

The loss in Theorem 5.3 comes from summing over all times up to a linear horizon. The actual certified blocks occupy a much smaller time support once the high-rank barrier width is allowed to shrink with the current rank.

Choose fixed high-rank data

$$a_0 < r_{hi} < 1, \quad 0 < \tau < r_{hi} - a_0, \quad (6.1)$$

and fixed low-rank data

$$a_0 < r_{lo} < 1, \quad 0 < \eta_{lo} < r_{lo} - a_0. \quad (6.2)$$

For constants $D, C_{sw} > 0$, put

$$S_M = \lceil C_{sw} \log(M+2) \rceil, \quad (6.3)$$

and, at a high parent rank $m \geq S_M$, certify with

$$\eta_{M,m} = \min \left\{ \tau, D \sqrt{\frac{\log(M+2)}{m}} \right\}. \quad (6.4)$$

Below the switch use the fixed tolerance η_{lo} . Every smaller tolerance reuses the fixed high-rank certification theorem at tolerance τ , so no rank-dependent startup assumption is introduced.

The nested first-passage proof remains unchanged. With $r_* = \min\{r_{hi}, r_{lo}\}$, every first-bad landing at threshold rank q is a direct first passage from the original shell and satisfies

$$E_{2^q}(n) < \frac{q+2}{r_*}. \quad (6.5)$$

Lemma 6.1 (duration corridor and feasible-time support)

There is a constant $K > 0$, depending only on the fixed parameters, such that for every outer shell I_M and every possible first-bad rank q , all cumulative first-passage times belong to a finite set $H_{M,q}$ satisfying

$$\# H_{M,q} \leq K \sqrt{(M+2) \log(M+2)}. \quad (6.6)$$

Proof

Consider one certified block from a parent shell I_m to threshold 2^q , and let h be its duration and t its active tolerance. The two deterministic orbit envelopes and the first-passage band give

$$(1-t)m - q \leq (1-a_0)h < (1+t)(m+1) - q + 1. \quad (6.7)$$

The width before division by $1-a_0$ is exactly $2tm + t + 2$. During the high phase, $tm \leq D \sqrt{m \log(M+2)} + \tau$, and the parent ranks decrease geometrically with ratio at most r_{hi} . Hence the sum of high-phase widths is $O(\sqrt{M \log(M+2)})$. The low phase begins below $S_M + 1 = O(\log M)$; summing its fixed-tolerance widths contributes only $O(S_M)$. Adding the integer rounding errors gives one interval of at most the size in (6.6) containing every feasible cumulative time. $\square$

The tagged-fiber proof also has the following support-sensitive form. If H is any finite set of positive times, $B \subseteq [1, Y]$, and the scaled reverse loss is at most D_q , then

$$\# \{n \in I_M : \begin{array}{l} \text{the first passage below } Y \text{ occurs at some } h \in H, \\ T^h(n) \in B, E_Y(n) \leq D_q \end{array} \} \leq \# H \left(1 + 3D_q \frac{2^M}{Y}\right) \# B. \quad (6.8)$$

This is Proposition 4.3 summed only over the declared time tags. No interval structure or density hypothesis on H is used.

Above the switch, choose C_{sw} large enough that the cap in (6.4) is inactive. The quadratic maximal-barrier estimate then has exponent $\eta_{M,m}^2 m = D^2 \log(M+2)$. Consequently the initial high failure and every high-rank landing target have a common density bound

$$d_{hi}(M) \ll (M+2)^{-p_{hi}}, \quad p_{hi} = \min\{C_{sw} \log 2, c_0 D^2\}, \quad (6.9)$$

where $c_0 > 0$ is the fixed quadratic maximal-barrier constant. The first term in the minimum pays for the dyadic endpoint boundary at the switch.

Theorem 6.2 (support-sensitive terminal profile)

Let $b_{lo} < b_{ent}(\eta_{lo})$ and let $0 < c_2 < \log 2$. For all sufficiently large M and $1 \leq L < S_M < M$,

$$\boxed{\frac{\# \text{Fail}_{M,L}^{sh}}{2^M} \ll d_{hi}(M) + \sqrt{M \log(M+2)} (M^2 d_{hi}(M) + e^{-b_{lo}L} + e^{-c_2 L})}. \quad (6.10)}$$

Outside this failure set, some iterate before

$$\frac{M}{1 - r_{hi}} + \frac{S_M + 1}{1 - r_{lo}}$$

lies below 2^L . If $\tau < \beta$, every iterate through the same witness is at most $n^{1+\beta}$, once M is sufficiently large.

Proof

Partition by the unique first failed certification rank. The direct-passage identity (6.5), Lemma 6.1, and (6.8) give, after division by 2^M , the time-support factor times $(q + 1)$ times the density of the rank- q bad landing target. Summing the high ranks contributes $O(\sqrt{M \log M} M^2 d_{hi}(M))$. The strict exponents $b_{lo} < b_{ent}(\eta_{lo})$ and $c_2 < \log 2$ absorb the linear rank factor in the low geometric tails and give the final two terms in (6.10). The initial failure contributes $d_{hi}(M)$.

The block ranks decrease geometrically, which gives the displayed clock. High blocks use tolerance at most τ . Every low block begins below $2^{S_M+1} = (M + 2)^{O(1)} = n^{o(1)}$, so its fixed-tolerance envelope is also below $n^{1+\beta}$ eventually. $\square$

Proof of Theorem 1.1

We prove the target in (1.4) by taking terminal rank $L_M = \lceil A \log_2(M + 2) \rceil$, while retaining the logarithmic clock and the $n^{1+\beta}$ orbit ceiling from Theorem 6.2.

As $\eta \uparrow 1 - a_0$,

$$\frac{1}{2} + \frac{\eta}{\log_2 3} \uparrow \log_3 2 = p_*.$$

Continuity of relative entropy gives

$$\sup_{\eta < 1 - a_0} b_{ent}(\eta) = D(p_* \| 1/2) = (1 - H_2(p_*)) \log 2. \quad (6.11)$$

Fix the strict parameters in (1.3). Choose

$$a_0 < r_{hi} < 1 - \frac{1}{c \log 2}, \quad 0 < \tau < \min\{r_{hi} - a_0, \beta, a_0\}. \quad (6.12)$$

Because $A > A_{FP}$, equation (6.11) permits fixed η_{lo}, b_{lo}, c_2 with

$$0 < b_{lo} < b_{ent}(\eta_{lo}), \quad 0 < c_2 < \log 2, \quad \frac{1}{2} < \frac{A \min\{b_{lo}, c_2\}}{\log 2}. \quad (6.13)$$

Choose $a_0 + \eta_{lo} < r_{lo} < 1$. Next choose D and then C_{sw} so large that $D/\sqrt{C_{sw}} \leq \tau$, $C_{sw} > A/\log 2$, and p_{hi} in (6.9) exceeds every fixed polynomial cost needed below.

Set

$$L_M = \lceil A \log_2(M + 2) \rceil. \quad (6.14)$$

Then $L_M < S_M$ eventually. From (6.10), every fixed

$$0 < \varkappa < \min\left\{p_{hi} - \frac{5}{2}, \frac{A \min\{b_{lo}, c_2\}}{\log 2} - \frac{1}{2}\right\} \quad (6.15)$$

is admissible after a small strict decrease to absorb powers of $\log(M + 2)$. The choices above make this interval nonempty. Moreover,

$$2^{L_M} \leq 2(M + 2)^A \ll_A (\log n)^A \quad (n \in I_M), \quad (6.16)$$

and

$$\frac{M}{1 - r_{hi}} + O(\log M) < cM \log 2 \leq c \log n \quad (6.17)$$

eventually.

For $2^J \leq X < 2^{J+1}$, shells below $J/2$ contain $O(X^{1/2})$ integers, while on the remaining shells $M^{-\varepsilon} \ll J^{-\varepsilon}$. Their dyadic sizes sum to $O(X)$, proving (1.4). Finally, (6.11) shows that the infimum of the strict exponents is exactly $1/(2(1 - H_2(p_*)))$; the endpoint is not claimed. $\square$

7. Quantitative companions

The two terminal profiles have different quantitative consumers. The support-sensitive profile gives the fixed-polylogarithmic headline, while the one-regime profile gives the sharper stretched-logarithmic exceptional rate and the clock consequences recorded here.

Proof of Theorem 1.2

We prove the target in (1.5) by choosing the terminal rank so that $2^{L_M} \leq \exp((\log n)^{1-\delta})$ on the source shell.

Fix $0 < \delta < 1$, put $\alpha = 1 - \delta$, and choose a one-regime pair (r, η) satisfying (5.1), with

$$r < 1 - \frac{1}{c \log 2}, \quad \eta < \beta. \quad (7.1)$$

Take

$$L_M = \lfloor \frac{(M \log 2)^\alpha}{\log 2} \rfloor. \quad (7.2)$$

The factor $M(L_M + 1)$ in (5.11) is absorbed by the exponential tail, so

$$\# \text{ Fail}_{M, L_M} \leq C 2^M e^{-c' M^\alpha}. \quad (7.3)$$

For $n \in I_M$,

$$2^{L_M} \leq \exp((M \log 2)^\alpha) \leq \exp((\log n)^\alpha).$$

The clock and path ceiling follow from (5.12) and (7.1). Lemma 2.1, with $\sigma = \alpha$, proves (1.5). $\square$

Raw-clock conversion

Let $s_h(x)$ count the odd states among $x, T(x), \dots, T^{h-1}(x)$. Direct induction gives

$$\text{Col}^{h+s_h(x)}(x) = T^h(x). \quad (7.4)$$

If $x \in W_\eta$ and $h \leq \lfloor \log_2 x \rfloor$, the positive main term in the affine iterate and the upper envelope in (3.8) give

$$s_h(x) \leq \frac{h}{2} + \frac{\eta \log x}{\log 3}. \quad (7.5)$$

Thus a one-regime chain has raw time at most

$$\frac{1}{1-r} \left(\frac{3}{2 \log 2} + \frac{\eta}{\log 3} \right) \log n + o(\log n). \quad (7.6)$$

For the two-regime chain, (7.6) applies to the high phase; the entire low phase has $O(S_M) = o(\log n)$ shortcut steps and at most twice as many raw steps. As $r \downarrow a_0$ and $\eta \downarrow 0$, the coefficient in (7.6) tends to

$$\frac{3}{\log(4/3)}.$$

The high parameters in (6.12) can therefore be chosen to satisfy any fixed shortcut and raw budgets strictly above their respective limits. To retain the literal raw-orbit ceiling in Corollary 1.3, run the argument with an internal exponent $0 < \beta' < \beta$. Every additional raw odd image is $3m + 1 = 2T(m)$, so $2n^{1+\beta'} \leq n^{1+\beta}$ eventually. This proves Corollary 1.3. $\square$

Fixed-power exceptional count

Fix $\alpha > 0$ and $0 < \sigma < 1$. Choose $0 < \delta < 1 - \sigma$. Then

$$\exp((\log n)^{1-\delta}) \leq n^\alpha$$

eventually, while $1 - \delta > \sigma$. Theorem 1.2 therefore implies (1.7), after decreasing the positive exponential constant and absorbing the finite startup.

Independent graded fixed-power clock

The second assertion of Corollary 1.4 is a fixed-depth companion, not an input to either headline theorem. We give the short derivation because it records a genuine time–descent tradeoff that the stronger polylogarithmic target alone does not express.

Put $g = 1 - a_0$. Fix $a_0 < r < 1$ and $0 < \eta < r - a_0$. For a certified source $x \in I_m$, set $Y_m = 2^{\lfloor r m \rfloor}$, let $G(x)$ be its first entrance into $(Y_m/2, Y_m]$, and put

$$H_m = \left\lceil \frac{(1 + \eta - r)m + 2 + \eta}{g} \right\rceil. \quad (7.7)$$

The strict inequality $\eta < r - a_0$ gives $H_m \leq m$ eventually. The upper half of the all-prefix envelope (3.8) then gives

$$T^{H_m}(x) < 2^{-gH_m + (1+\eta)(m+1)} \leq 2^{rm-1} \leq Y_m. \quad (7.8)$$

Hence the first passage defining G occurs by time H_m , and

$$H_m \leq \frac{1 + \eta - r}{g} m + O_{r,\eta}(1). \quad (7.9)$$

We record why a fixed number of these stages retains natural density one. After absorbing finitely many startup shells, let $\tilde{W}_\eta$ denote the certified set and put $P(S) = \tilde{W}_\eta \cap G^{-1}(S)$. If S is (C, D) -dense and $0 < \chi < r$, then $P(S)$ is $(C', \chi D)$ -dense for every sufficiently small fixed $D > 0$. Indeed, for every first passage of length at most

H_m , the loss in (4.5) satisfies $E_{Y_m} \leq H_m/2$, and $H_m/(2Y_m) \leq 1/3$ eventually. Apply Proposition 4.3 with this loss budget and with $B = (Y_m/2, Y_m] \setminus S$. Since $|B| \leq CY_m^{1-D}$, the transported part of the shell complement is

$$\ll Cm^2 \frac{2^m}{Y_m} Y_m^{1-D} \ll Cm^2 2^{(1-rD)m}. \quad (7.10)$$

For fixed $0 < \chi < r$, the polynomial factor is absorbed by $2^{(r-\chi)Dm}$, uniformly at cost $O_{r,\chi}(D^{-2})$. Proposition 3.3 absorbs the certification complement after D is restricted below a fixed positive threshold. Dyadic summation proves the claimed density update. Iterating it any fixed number of times therefore still gives a natural-density-one set.

Now fix $0 < \alpha < 1$ and $\varepsilon > 0$. Choose $0 < \alpha' < \alpha$ with $c_*(\alpha - \alpha') < \varepsilon/3$, then choose a fixed $R \geq 1$ and $r = (\alpha')^{1/R} > a_0$. Finally choose $0 < \eta < r - a_0$ so that $c_*(1 - \alpha')\eta/(1 - r) < \varepsilon/3$. On the resulting R -fold pullback set, write $n_0 = n$ and $n_{i+1} = G(n_i)$. Finite startup changes only a constant, so $n_i \leq K_* n^{r^i}$ for a fixed K_* , and consequently $n_R \leq K_* n^{\alpha'} \leq n^\alpha$ eventually. Summing (7.9) gives

$$\begin{aligned} k_R &\leq \frac{1 + \eta - r}{g} \frac{1 - r^R}{1 - r} \log_2 n + O_{\alpha,\varepsilon}(1) \\ &= c_*(1 - \alpha') \left(1 + \frac{\eta}{1 - r}\right) \log n + O_{\alpha,\varepsilon}(1). \end{aligned} \quad (7.11)$$

The two strict parameter margins leave a final third of the clock slack to absorb the fixed remainder, proving (1.8).

8. Scope

The argument proves an orbit-minimum statement on a set of natural density one. It does not prove descent for every starting value, exclude nontrivial cycles, or control the orbit after the selected witness. The endpoint $\delta = 1$ is a strict parameter limit and is not claimed. No finite computation is used as an all-depth premise.

The proofs of Theorems 1.1 and 1.2 use only the optimized first-passage chain: the parity-vector bijection, maximal-barrier estimate, first-passage reversal, exact reverse-product loss, nested direct re-certification, and compressed feasible-time support. These proofs use no fixed-time endpoint-fiber moment, stochastic independence between orbit blocks, mixing statement, Diophantine reduction, or generated-target equidistribution hypothesis. The Boolean walk in Lemma 3.1 is exact counting over the parity words supplied by Proposition 2.2, not a stochastic model of an individual Collatz orbit. The independent graded-clock companion uses only a fixed finite dense-set pullback and is not on either headline dependency chain.

Research and software disclosure

Generative-AI systems were used in proof exploration, adversarial auditing, finite diagnostics, formalization, and exposition. The author selected the theorem and proof architecture, reviewed the resulting artifacts, and accepts responsibility for the manuscript. Finite diagnostics are supporting evidence only and are not premises of any theorem.

The separate Lean package formalizes the same optimized chain, from loss-filtered transport through the natural-density assembly, and separately formalizes the fixed-depth graded-clock companion. Its public `Main` theorem has the strict ranges $A > A_{FP}$, $c > c_*$, and $\beta > 0$ from Theorem 1.1; its landing inequality is strict and hence slightly stronger than the manuscript's weak inequality. The full build, dependency report, placeholder scan, and public-root axiom audit pass, with logical dependencies `propext`, `Classical.choice`, and `Quot.sound`. These checks supplement rather than replace the written proof.

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